Exact-span two-of-four consensus

CS781 Phase 2 legal named entity recognition | source-audited report

Task and evidence

Predict exact original-character entity spans in 12 legal categories. The source archive holds the actual code, configuration, run record, selected weights, and prediction ZIP. This report was produced after training from those records; it is not an executed historical notebook.

Architecture

This is a prediction-level ensemble, not a newly trained neural architecture. A span survives if at least two of the first four models output the same start, end, and label; conflicting overlaps are resolved by vote count then span length.

Path	LegalSpan -> B1 -> C1 -> A0
Output	Exact span votes -> Overlap resolution -> Prediction JSON

Training and validation

Method: Two-of-four exact span vote. This export's archived held-out strict macro-F1 is 0.463067 on the 12-document local validation split. Its selected checkpoint and prediction ZIP hashes are in evidence/run_summary.json. The official test set has 35 document IDs; no hidden Codabench score is established by this archive.

Implementation map

Key original files: scripts/build_consensus.py, lner/data.py. The companion notebook explains the tensor route and reads archived evidence without presenting invented execution outputs.

Reproducibility boundary

The package includes exact source snapshots and small run records, plus the archived prediction ZIP. Large model weights, tokenizers, and course data stay in the separate complete archive. Original machine paths in config files need remapping for a new environment. The consensus rule was selected on validation; 100M/200M D222 exports were completed after the original deadline where marked.

Interpretation

Model variants change either pretrained backbone, parameter-update method, NER head, or pretrained checkpoint. A score difference does not isolate one factor unless all other choices match. The local held-out set is small, so its ranking should not be described as a hidden-test result.

Recorded model and adaptation

The source and run records together identify the following settings. Counts include the Phase 2 NER head. The method changes which backbone parameters are updated; it does not change the original Phase 1 tokenizer during this run.

Field	Archived value
Base family	Ensemble
Adaptation	Two-of-four exact span vote
Total parameters	see source models
Trainable parameters	see source models
Backbone LR	see original config
Head LR	see original config
Pretrained SHA-256	see run.json…
Selected NER SHA-256	not applicable…

Training signal and exact-span evaluation

For the common NER runner, the character tagger minimizes weighted cross-entropy over 25 BIO categories; A0 also matches its verified B1 teacher with temperature-2 KL. D222 adds CRF negative log-likelihood and a weighted boundary objective. HeaderCRF combines cross-entropy, CRF, and boundary losses. The validation scorer counts a prediction as correct only when the start, end, and label all match the gold span; macro-F1 averages the 12 per-label F1 values equally.

Label-level held-out scores

The following F1 values come from the archived selected validation prediction. Low-frequency labels are sensitive to a few documents, so read them beside their support and error counts in evidence/selection.json.

Entity label	F1	TP	FP	FN
APP	0.070	5	29	104
RESP	0.033	2	20	99
A.COUNSEL	0.525	21	14	24
R.COUNSEL	0.411	15	27	16
JUDGE	0.582	16	4	19
WIT	0.000	0	2	21
AUTH	0.430	96	146	109
COURT	0.864	184	32	26
STAT	0.651	205	81	139
PREC	0.312	67	100	195
DATE	0.967	321	11	11
CASENO	0.713	92	28	46

Verification and reproducibility

The prediction ZIP has a single root prediction.json, passed CRC, and contains exactly 35 official test IDs. Its SHA-256 was compared with the completed run record. Each copied code file has a SHA-256 listed in evidence/SOURCE_HASHES.json. Exact re-execution requires the original selected weights, tokenizer and course data preserved in the complete archive; this compact package does not embed them or assert that historical notebook output cells existed.